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Critical review of data-driven decision-making in bridge operation and maintenance

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ABSTRACT

Bridges are critical infrastructure, and effective operation and maintenance (O&M) is essential for ensuring the good condition of bridges. Owing to the increasing complexity of modern bridges and the availability of information technologies (e.g. sensors, laser scanners, and ultrasonic radar) for the collection of massive data, bridge O&M and decision-making gradually shift toward a data-driven manner. However, both the bridge industry and the academia still do not have a common understanding of the latest progress, challenges, and trends of data-driven bridge O&M decision-making. Thus, through a critical review of 485 articles, this paper investigates current data-driven bridge O&M decision-making in detail, including mainstream data types, issues related to data management, and typical application areas using these data. Challenges to implement data-driven bridge O&M decision-making are identified, such as lack of standard data needs, lack of data integration, and lack of standard procedures. Future research opportunities to address the challenges are proposed. This paper can help bridge O&M teams by identifying suitable data and applications to make informed decisions that align well with their needs meanwhile serve as a basis for future research efforts in this area.

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1. Introduction

Bridges are critical transportation infrastructure, and many bridges are aging quickly due to multiple reasons, e.g. increasing traffic. Therefore, it is important to ensure the good condition of bridges through informed operation and maintenance (O&M) decisions (Frangopol & Bocchini, 2012). Traditional bridge O&M depends on experience of engineers, which can be subjective and hysteretic. Given the increasing complexity of bridges and availability of information and communication technologies (ICTs), bridge O&M decision-making is shifting towards a data-driven manner (Auckland University, 2015; Helmerich, Kuhn, & Nussbaumer, 2007; Woldeesenbet, 2014). For one thing, sensing techniques, e.g. sensors and laser scanners, can collect rich data to accurately reflect bridge conditions and serve as a basis for decision-making; for another, analytical methods, e.g. mechanical models, machine learning (ML) models, and multi-objective optimisation algorithms, are developed to analyse data, evaluate structure conditions, and support decision-making. It is found 75% bridge O&M participants believe that data are the key for successful bridge O&M, while one knowledge gap is how to effectively collect and utilise data to make decisions (Woldeesenbet, 2014).

Similar to other industries, advances of researches can drive the transformation of bridge O&M practices (Wu, Xu, Mao, & Li, 2017; Wu, Wang, & Wang, 2016). However,

both the industry and academia of bridge sector do not have a common understanding of the latest progress, challenges, and trends of studies of data-driven O&M decision-making. Some studies review O&M of transportation infrastructure, focusing on decision-making optimisation (Frangopol & Bocchini, 2012; Kabir, Sadiq, & Tesfamariam, 2014; Mardani, Zavadskas, Khalifah, Jusoh, & Nor, 2015). These studies review details of optimisation algorithms, but do not cover how to collect, process, and manage data that these algorithms need (Auckland University, 2015). Meanwhile, some studies review methods that collect and manage data, e.g. computer vision (Koch et al., 2014), robotics (La, Gucunski, Kee, & Nguyen, 2015; Liu et al., 2014), and sensors network (Yun, Cho, Park, Min, & Park, 2014), BIM (Bradley, Li, Lark, & Dunn, 2016; Costin, Adibfar, Hu, & Chen, 2018), and Civil Information Model (CIM) (Sankaran et al., 2016). These studies focus on technical details and applications of technologies in the entire transportation sector (e.g. highway, bridge, and tunnel) and different phases (e.g. design, construction, and O&M). Thus, they are not specific enough for bridge O&M and guide transformation to data-driven decision-making in practice.

To address the limitation, this paper provides a review of researches that: 1) includes 485 journal articles to reveal the latest progress, challenges, and trends of data-driven bridge O&M decision-making, focusing on data, data management,

and critical applications; and 2) identifies challenges and research opportunities. This review also covers manuals, standards and reports of bridge O&M published by transportation departments and highway and road authorities in developed countries to to-some-extent relate researches to current practices.

2. Review approach

The approach proposed by Thome, Ceryno, Scavarda, and Remmen (2016) was used to identify and review articles. The first step defines the review scope as data-driven decision-making for bridge O&M. The second step decides the database and keywords for searching. The Web of Science database was selected due to wide coverage and high quality (Bradley et al., 2016). Data-driven decision-making includes data collection, analysis, storage, sharing, and decision-making (Woldesenbet, 2014). Therefore, the query was (*bridge**) AND (*data NEAR/5 manage* OR information NEAR/5 manage* OR data NEAR/2 collect* OR information NEAR/2 shar* OR data NEAR/2 exchang* OR decision* OR decision* NEAR/2 make* OR optimi* OR multi* NEAR/2 object* OR multi* NEAR/2 criter**) AND (*planning* OR monitoring* OR maintenance* OR inspect* OR repair* OR rehabilitat**).

The query includes the review scope, key processes of data-driven decision-making, and O&M tasks. It also considers words with similar meanings (e.g. data and information), and common variations and combination of words (e.g. exchang* can include exchange and exchanging, and NEAR/n specifies the number of words between two terms). The search returned 766 articles which were filtered in the third step. First, only peer-reviewed journal articles were included to ensure quality. Then, the abstract and keywords of each article were checked to ensure that they can fit in the scope. As a result, 485 articles were left for reviewing. First, general information (e.g. publication year, journal, and region) was investigated to reveal an overall picture. Then, a content analysis, using the codes in Table 1, was applied to extract main ideas of each article.

3. Overview of the results

Studies on data-driven decision-making in the infrastructure sector began in the 1980s. However, early studies may not fit in the review scope (e.g. they are not directly related to bridge O&M or they are not published in the peer-reviewed journals archived in the WoS). As shown in Figure 1, there is a stable increase of publications overtime, except the sharp increase in 2017 and the decrease in 2019. The increase is probably due to that limited budgets and lack of data for decision-making are long-existing challenges for bridge O&M departments. Thus, advanced ICTs and analytic methods are consistently and increasingly needed to improve O&M. Besides, modern hardware and software provide more computing capability to support algorithms and programs that were difficult to run, which also contribute to the increasing number of publications.

There are 51 journals that have published relevant articles. Figure 2 illustrates journals with more than five publications. *Structure and Infrastructure Engineering* is the most significant journal with 67 publications, followed by *Journal of Bridge Engineering* and *Engineering Structures*, with 42 and 37 publications, respectively.

Finally, as shown in Figure 3, the articles are produced in eight regions. The US publishes approximately 42% articles, indicating a dominant role in the area. Despite of the small number of publications, other regions also conduct important researches are relatively less conducted in the US, such as the researches of historical and masonry bridges monitoring in Europe (Pachón, Castro, García-Macías, Compan, & Puertas, 2018), the expert system based bridge management systems (BMS) in Canada and Japan (Hawk & Small, 1998; Miyamoto & Asano, 2018), the ontology-based BMS in the Netherland (Luiten, Böhms, Alsem, & O'Keeffe, 2019; NRA, 2018), and the sensors layout optimisation in China (Zhou, Xie, Yi, & Li, 2019).

4. Data for data-driven bridge O&M decision-making

4.1. Data types and collection

Data are qualitative or quantitative descriptions of a bridge, its components, and the surrounding environment. Table 2 summarises main data types (meaningful data) for data-driven bridge O&M applications, which are divided at bridge and network level. Meaningful data (also called information) are converted from raw data, where useless and random raw data are processed and assigned with meaning under specific contexts (London DoT, 2013). In a data-driven manner, raw data are collected by sensors or non-destructive technologies (NDTs).

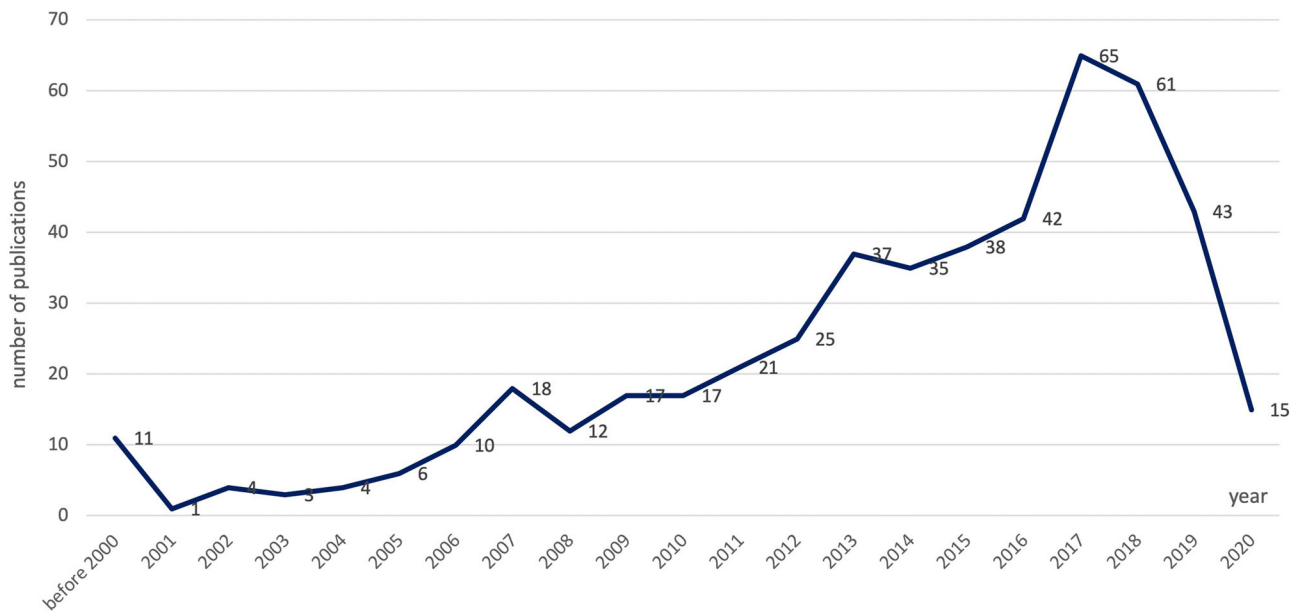
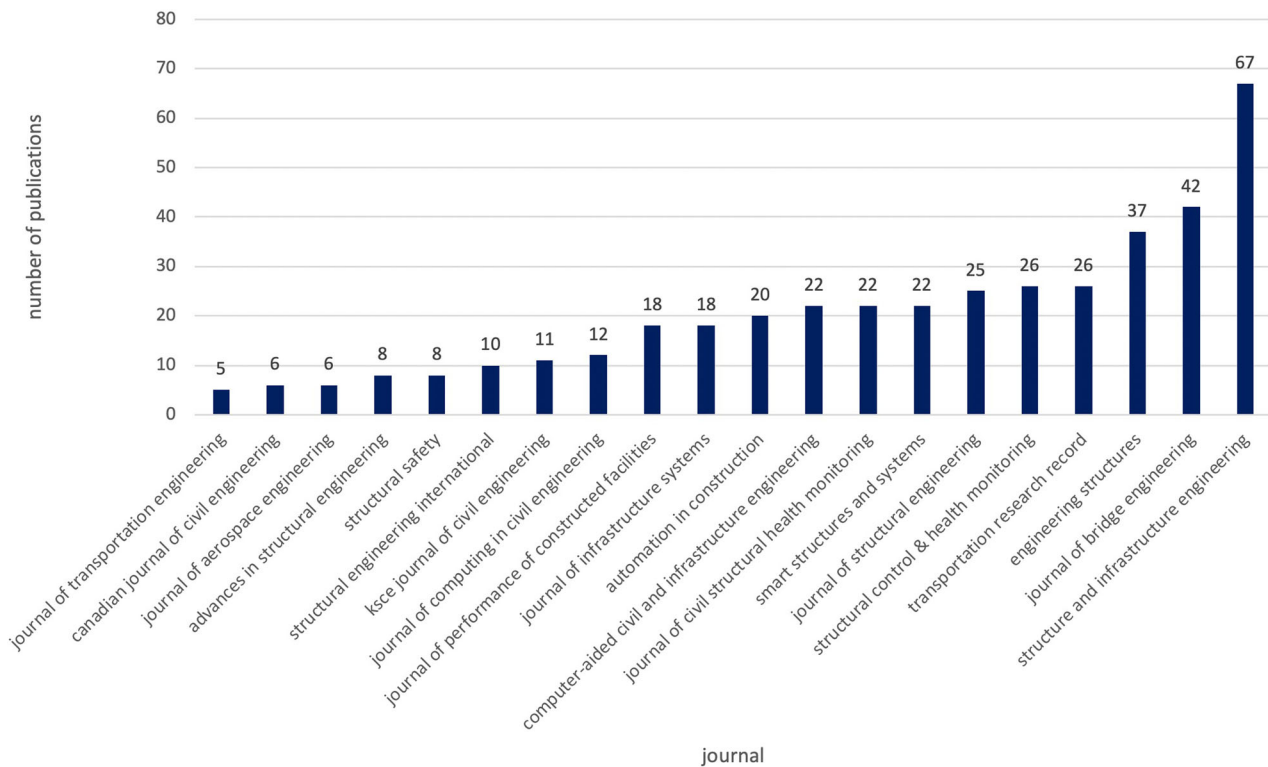
Bridge profile is the most common meaningful data, followed by structure responses, traffic data, and damage data. Data of structure responses and damages are mostly derived from raw data, whereas other data can be obtained from second-hand sources. For instance, bridge profile and IMR&R data can be collected from project documents and inspection reports, respectively, and traffic, environment, and social data can be collected from databases of relevant agencies. Second-hand sources are important for O&M researches, because they provide data which are hard to collect using sensors and NDTs, e.g. maintenance cost, or data which are hard to access because they are managed by external stakeholders, e.g. traffic data.

4.1.1. Sensor based data

As shown in Table 3, data collected by sensors are divided into three groups. For structural data, acceleration, strain, and stress are the most common types because they are the basis for most structure analyses (Miyamoto & Yabe, 2012). Displacement and deflection mainly include vertical and horizontal displacement (Costa & Figueiras, 2013; Koo, Brownjohn, List, & Cole, 2013), rotation (Costa & Figueiras, 2013; Goulet, Kripakaran, & Smith, 2010), and incline

Table 1. Categories and codes of article texts.

Categories	Codes
Bridge O&M data	(1) data types, (2) data collection methods
Application areas	(1) structure related knowledge investigation, (2) failure probability and life expectancy evaluation, (3) structure condition indexes evaluation, (4) decision-making of inspection maintenance, repair, and rehabilitation (IMR&R) tasks planning, (5) decision-making of O&M tools selection
Data management	(1) data needs and quality, (2) data storage and organisation, (3) data sharing

**Figure 1.** Distribution of articles overtime (2020 not complete year).**Figure 2.** Distribution of articles by journal.

(Chiu, Lin, Hung, Sung, & Chang, 2014; Santos, Cremona, Orcesi, & Silveira, 2017). Displacement data are also essential for damage detection but are less collected, possibly due

to additional preparation procedures, e.g. setting reference points, that are often required. For environment data, temperature and wind data are widely collected due to their

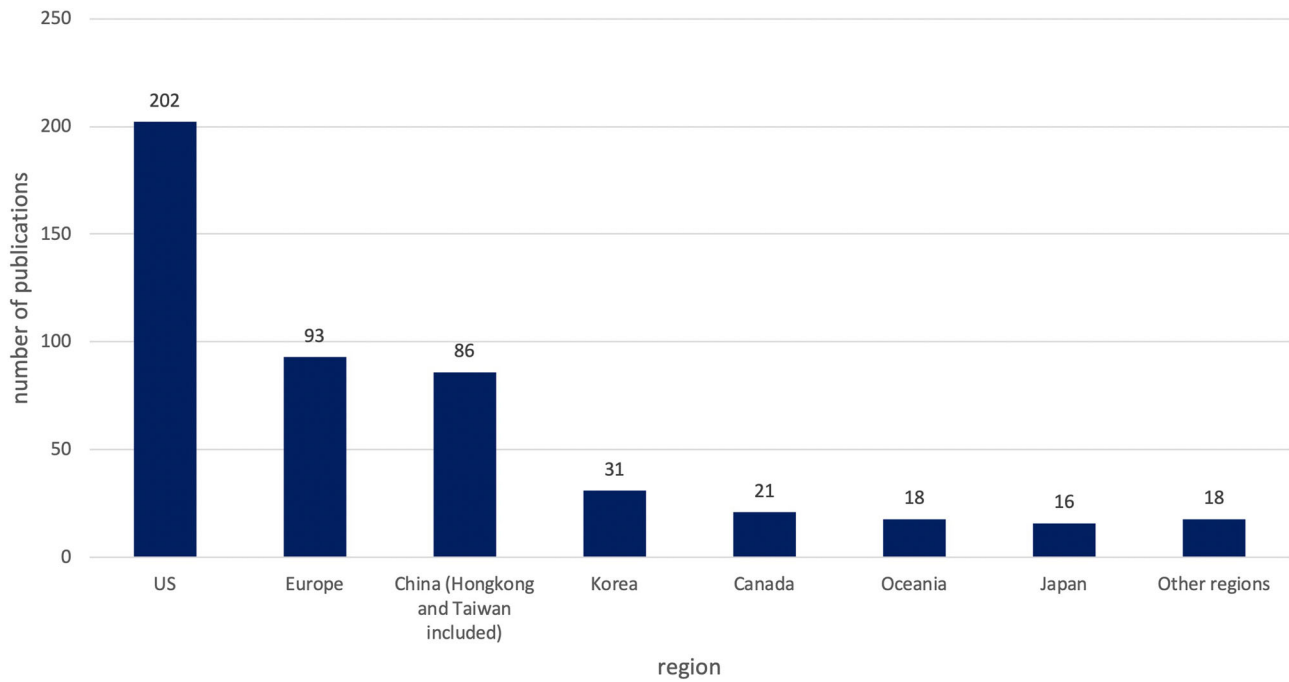


Figure 3. Distribution of articles by region.

Table 2. Main data types used for data-driven bridge O&M applications.

Level	Data type	Counts	Instances
Bridge	Bridge profile	278	Age, location, type, properties of components (e.g. geometry, number, material properties (e.g. comprehensive stress and deterioration rate), and connection of components), structure conditions
	IMR&R data	119	IMR&R history (e.g. duration, cost, budget, and recorded effect of tasks), costs of SHM system and NDTs
	Damage and defect data	129	Location and geometry of damages (e.g. length, width, and orientation of cracks, area of cracks, stain, spalling, delamination, and inadequate lubrication, and size and depth of scour, spalling and porthole), and corrosion initiation time and propagation rate
	Structure responses	193	Statistics of structure responses (e.g. displacement, vibration, stress and strain, and seismic load)
	Network Traffic data	132	Statistics of traffic volume (e.g. annual average daily traffic (AADT), average daily traffic (ADT), and truck traffic), detour distance and duration, vehicle type, speed, weight, and the number of axles
	Environment data	114	Statistics of temperature, wind, water flow, chloride concentration, diffusion, and density, CO2 emission, freeze-thaw, ice, moisture, and rain volume
Network	Social data	74	Discount rate, driver income, traveller cost, accident rates and fatality costs

close relationships with structures (Hiasa, Catbas, Matsumoto, & Mitani, 2016; Webb, Vardanega, Fidler, & Middleton, 2014). For traffic data, the focus is on collecting traffic load, especially the load of heavy trucks that can cause damages on the structure (Lou, Nassif, Su, & Truban,

2017; Radovic, Ghonima, & Schumacher, 2017; Song et al., 2019).

Table 3 also presents mainstream sensors to collect data. A structure health monitoring (SHM) system, i.e. mounting sensors on the structure that often forms a network, is the dominant method since it can deploy sensors for different data types (FHWA, 2014). SHM systems are generally installed on superstructures in practice. However, efforts are also made to use sensors to monitor more bridge parts, e.g. movable parts (Catbas et al., 2014) and O&M components, e.g. fibre-reinforced polymer (FRP) (Du, Li, & Li, 2019; Yang, Frangopol, & Teng, 2019).

Sensors network can be wired or wireless. The key difference is that in wireless sensors network (WSN), sensors work independently while data transferring is based on wireless communication techniques rather than cables. It is recognised that WSN is easier to maintain, more stable, and more cost-effective in the long-term (Yun et al., 2014; Zhou et al., 2019). However, in practice, the wired system is still implemented more widely because: 1) the costs (especially the initial investment) of wired network are lower; and 2) in real projects, WSN faces several technical challenges that can affect its reliability and data quality, e.g. the lack of power supply, distributed sensors control, and unstable sensor communication (FHWA, 2014). Hence, WSN is often deployed on critical bridges (e.g. river crosses) in practice, and efforts are continuously made to address these challenges (Section 4.2). Besides, mobile sensors, i.e. installing sensors on vehicles or using sensors in smart phones to collect data, are gaining increasing attention and have been applied in practice as an economic alternative to WSN. However, mobile sensors can only monitor superstructure acceleration (Sitton, Rajan, & Story, 2020).

To derive meaningful data from raw data, electronic signals are first converted to digital signals (Agdas, Rice,

Table 3. Sensor based raw data collection.

Type	Signal data	Mainstream sensors	Counts
Structure responses	Strain and stress	Strain gauges (Goulet et al., 2010; Whelan, Gangone, Janoyan, & Jha, 2009), pressure plates and transducers (Santos et al., 2017; Zhou et al., 2017), fibre-optic sensors (Zonta et al., 2014), Fibre Bragg grating (FBG) sensors (Carrion et al., 2017a; Li et al., 2017)	100
	Acceleration and vibration	Accelerators and accelerometers (Boddupalli, Sadhu, Azar, & Pattysan, 2019; Dragos et al., 2018; Pan et al., 2018), GPS (Xi, Chen, Meng, Jiang, & Chen, 2018), GNSS (Moschas & Stiros, 2015)	115
	Displacement and deflection	Inclinometers (Hoult et al., 2010; Santos et al., 2017), tilt-meters (Goulet et al., 2010), linear variable differential transformers (Chiu et al., 2014; Costa & Figueiras, 2013), linear potentiometric displacement transducers and linear variable doppler transformers (LDV) (Hoult et al., 2010; Webb et al., 2014), GPS (Sun & Sun, 2018; Xi et al., 2018)	63
Environment data	Temperature	Resistance temperature detectors (Costa & Figueiras, 2013), thermometers (Chiu et al., 2014), thermistors (Jeong et al., 2017; Jeong et al., 2016)	50
	Wind speed	Anemometers (Jang et al., 2010; Koo et al., 2013; Sun & Sun, 2018)	23
	Water and humidity	Climate stations (Costa & Figueiras, 2013; Yu & Ou, 2017), relative humidity meters (Hoult et al., 2010; Jang et al., 2010; Webb et al., 2014)	16
Traffic data	Traffic volume	FBG sensors (Chen, Wu, & Feng, 2019), weight-in-motion systems (Carrion et al., 2017a; Koo et al., 2013)	10
	Vehicle load		19
	Vehicle speed		8

Martinez, & Lasa, 2016). Then, various methods can be applied (Table 4), which reduce noise, save computation costs, and assist data interpretation, e.g. extracting modal properties (i.e. natural frequency, damping ratio, and mode shape). It should be noted: 1) After noise reduction, the output structure responses, e.g. the statistics of temperature and vehicle load, can already be meaningful for certain applications (Xia, Cai, Pan, & Yu, 2016), and other methods can be applied selectively according to situations listed in Table 4. 2) Some data can be used to compute or combined with other data, using mechanical models addressing their relationships. For instance, traffic load data can be converted to structure strain and stress (Ma, Xu, Wang, & Chen, 2018), and displacement and strain data can be fused together to gain more comprehensive monitoring results (Soman, Kyriakides, Onoufriou, & Ostachowicz, 2018).

4.1.2. Ndts based data

The major data types collected by NDTs are shown in Table 5. Photos are the most common data, followed by radar data, acoustics data, and infrared images. This complies with the application of NDTs in practice in countries like the US (Modjeski & Masters Inc, 2017). Although point clouds can

accurately capture structure features, it is only adopted by seven articles. This is possibly due to the strict requirements on equipment, environment, and operators to generate point clouds (Koch et al., 2014). Thus, more studies choose to create point clouds from photos (Khaloo, Lattanzi, Cunningham, Dell'Andrea, & Riley, 2018; Morgenthal et al., 2019). Point clouds and photos can capture both local features, e.g. girder cracks, and global features, e.g. holistic bridge geometry. In contrast, acoustic, radar, and infrared data often capture local features only, which however, can reflect both surface and sub-surface conditions, whereas photos and point clouds can only reflect surface conditions (Abu Dabous, Yaghi, Alkass, & Moselhi, 2017; Gucunski, Kee, La, Basily, & Maher, 2015).

Mainstream data collection devices are also listed in Table 5. In practice, data collection using NDTs is an iterative process, where routine inspection (usually visual inspection) is periodically performed, while in-depth inspection and damage inspection are performed when suspicious damages are identified in routine inspection or when damages happen, respectively (FHWA, 2014; MLIT, 2015). Ad-hoc inspection can also be performed after unusual events (e.g. earthquake or flood) (NCHRP, 2007). Besides, element-wise inspection has been adopted by more DoTs, which requires inspection of each bridge elements (e.g. a component and a

Table 4. Conversion of sensor collected raw data.

Purpose	Mainstream methods	Applicable situation
Reduce noise	Removing noise: pick-peaking (Navabian & Beskhyroun, 2019; Whelan et al., 2009), high-low cut-off pass filter, band-pass filter (Chiu et al., 2014; Sun et al., 2016) and Kalman filter (Xi et al., 2018) Smoothing values: autoregressive moving averaging and time averaging (Miyamoto & Yabe, 2012; Yu & Ou, 2017)	Required in most situations
Save computation cost	Sampling (Jang et al., 2010), compress sensing (Huang, Beck, Wu, & Li, 2014), clustering (Vanniamparambil et al., 2013), principal components analysis (PCA) (Li et al., 2017; Santos et al., 2017), and singular value decomposition (Zhou et al., 2017)	The data volume is large
Data transformation and system identification	Frequency domain decomposition, wavelet decomposition and transformation (Yu & Ou, 2017; Zhou et al., 2017), Fourier transform (Gucunski et al., 2015), and Hilbert-Huang and Teager-Huang transforms (Pan et al., 2018; Zhou et al., 2017), second order blind identification, eigensystem realization algorithm, natural excitation technique (Kvåle, Øiseth, & Rønnquist, 2017), and stochastic subspace identification (Ubertini, Gentile, & Materazzi, 2013)	Modal properties are needed

Table 5. NDTs based data collection.

Type	Description	Devices	Counts
Point clouds	Spatial points generated by projecting light bars on objects	Terrestrial laser scanner (TLS) (Riveiro et al., 2016; Turkan et al., 2018), Light Detection and Ranging (LiDAR) (Dai et al., 2014; Khaloo et al., 2018)	7
Photo images	Photographs taken by cameras	Hand-held cameras, smart phones, and tablets (Khaloo et al., 2018)	32
Acoustics and ultrasonic	Reflected sound waves	Impact echo (Vaghefi et al., 2012), acoustic emission (AE) devices (Li et al., 2019; Zárate et al., 2012), ultrasonic devices (Azari, Gucunski, Kim, & Dinh, 2016)	12
Radar	Reflected electric and magnetic pluses	Microwave (Budelmann, Holst, & Wichmann, 2014), sonar (Hayden & Puleo, 2011), eddy current (Soliman et al., 2013), and ground penetration radar (GPR) (FHWA, 2018)	14
Electrical and chemical	Electrical and chemical properties	Electrochemical chloride sensors and coil sensors (Budelmann et al., 2014), electrical resistivity devices (Gucunski et al., 2015)	9
Infrared images	Temperature and radiation properties of structures	Infrared thermography cameras (Hiasa, Birgul, & Catbas, 2017; Terzioglu, Karthik, Hurlebaus, & Hueste, 2019)	11

group of components) (AASHTO, 2010). Modern NDTs can be applied to inspect elements not covered by traditional inspection manuals, e.g. cables systems and mechanical components of movable bridges (TxDOT, 2020). However, inspection can still be inefficient as it requires sending staff onsite to collect data (Modjeski & Masters Inc, 2017).

Hence, many studies are conducted to address the limitation. First, equipment, such as TLS (Riveiro, DeJong, & Conde, 2016), can be fixed on or next to a bridge for real-time monitoring, but it is usually restricted by a blocked sightline between devices and targets (Koch et al., 2014). Thus, the Mobile Mapping System (MMS) is gaining popularity. MMS generally includes land vehicles and Unmanned

Aerial Vehicles (UAVs) (Ham, Han, Lin, & Golparvar-Fard, 2016). Land MMS refers to vehicles mounted by NDTs, e.g. cameras and TLS, which move through a bridge to collect data. However, the collection can be slow and blocked by barriers (Koch et al., 2014).

Recently, UAVs are increasingly applied since they can reduce data collection time significantly and access most blind spots (Boller, Starke, Dobmann, Kuo, & Kuo, 2015). However, UAVs require additional calibration, path planning, and control. Besides, the data collected are often less accurate than data collected by fixed equipment due to the limited carrying capacity and vulnerability to environment (e.g. wind) of UAVs (Liu et al., 2014). Finally, based on the

Table 6. Conversion of NDTs collected data.

Data type	Filtering methods	Analysing methods	Derived meaningful data
Images	Binary transformation, normalisation, clustering, slide-window, Gaussian filter, band-pass filter, and grey-scale transformation (Hiasa et al., 2017; Winkler & Hansen, 2018; Yeum & Dyke, 2015)	Detect damages: edge detection (Gul, Catbas, & Hattori, 2015; Yeum & Dyke, 2015), digital image correction (Winkler & Hansen, 2018), support vector machine (SVM) (O'Byrne et al., 2013), artificial neural network (ANN) (Gul et al., 2015; Turkan et al., 2018) Refine images or generate point clouds: photo-grammetry or video-grammetry (Boller et al., 2015; Khaloo et al., 2018; Morgenthal et al., 2019)	Surface damages, e.g. spalling, crack, stain, delamination Refined, integrated, or stitched images, and point clouds (usually used to improve algorithms that detect damage and components)
Point clouds	Octree filtering, PCA, and voxelization, (Riveiro et al., 2016), multi-view stereo and semi-global matching (Khaloo et al., 2018; Morgenthal et al., 2019)	Identify components: clustering, ANN, and rules-based reasoning (Koch et al., 2014) Detect damages: comparing dense point clouds (as-built model) with as-design model (Vaghefi, Ahlborn, Harris, & Brooks, 2015)	3D models of bridge components Surface damages, e.g. portholes and spalling
Acoustics, radar, and electrical and chemical data	Band-pass filter, time averaging, and time-zero correction (Vanniamparambil et al., 2013)	Mechanical models that analyse attributes of reflected pulses, e.g. frequency and strength (Zárate et al., 2012), abnormality detection (Li et al., 2019; Vanniamparambil et al., 2013)	Both surface and sub-surface damages, e.g. delamination areas and cracks

raw data collected by NDTs, some bridge profile and damage data can be derived using methods illustrated in Table 6.

4.2. Data needs and quality

Data needs and quality are important for data-driven O&M, which define what data are required for certain applications and to what degree of detail. Currently, most data needs are defined in bridge inspection manuals and are mainly for data collection. For instance, primary inspection only requires structure photos and general evaluation and description of damages, and detailed inspection can require accurate measurement of components' structural performance (e.g. concrete strength) and surface and subsurface damages (Mainroads, 2013; TxDOT, 2020). A few studies also attempt to define data needs for O&M applications, e.g. defining the data needs for bridge life estimation (Helmerich et al., 2007), developing an information delivery manual to transfer inspection data into in BIM or BMS (Sacks et al., 2018), and investigating information requirements of stakeholders (e.g. the owner and maintenance teams) (Shim, Kang, Dang, & Lee, 2017).

In addition, many efforts have been made to increase the quality of collected data. For SHM systems, both researchers and DoTs are implementing more advanced sensors to reduce noise. For instance, fibre-optic sensors are increasingly applied because they are immune to changes of temperature and electric/magnetic field (Woldesenbet, 2014). Enhanced energy management can also reduce the possibility of sensor failure. For instance, the threshold-based monitoring is usually mentioned, i.e. most sensors remain inactive until structure responses monitored by a few sensors exceed certain thresholds (FHWA, 2014).

Data transferring also affects data quality. Hence, studies propose procedures to improve data transferring in the network (e.g. the multi-hop transmission) and the energy-effective procedures to transfer data remotely (i.e. setting base-stations to temporarily store data onsite then transfer data to remote databases periodically) (Dragos, Theiler, Magalhaes, Moutinho, & Smarsly, 2018; Zhou et al., 2019). Studies are also conducted to support data fusion and synchronization between sensors, taking use of the relationships of different structure responses and modal properties (Dragos et al., 2018; Soman et al., 2018). Besides, studies have identified various affecting factors of data quality, e.g. data volume, monitoring time, temperature, humidity, and wind, which can explain data variance and help proposing improved monitoring plans (Arjomandi & Araki, 2019; Chiaia, Marasco, Ventura, & Quirini, 2020).

Another trend is to optimise sensors layout (e.g. the number, location, and orientation of sensors), considering various affecting factors, e.g. energy consumption, information accuracy and efficiency, transmission distance, sensors independency and coverage, and correlations between monitored components. Current methods to find the optimal sensors layout include the effective independence method, the kinetic energy method, and multi-objectives optimisation methods, e.g. Genetic Algorithm (GA), Firefly Algorithm, and Swarm Particle Algorithm (Sunca, Okur, Altunışık, & Kahya, 2020; Wang, Yuan, Wu, & Wang, 2019; Zhou et al., 2019).

For NDTs, practices to improve data quality focus on operational management, e.g. establishing quality assurance policy and checking personnel qualification (NCHRP, 2007; TxDOT, 2020). As it can be inefficient to access relevant data for inspection (e.g. the accurate size and historical conditions of components), portable devices (e.g. tablets) have

been widely applied by DoTs in Finland and Germany to provide inspectors real-time access to remote databases and 3D models (Modjeski & Masters Inc, 2017). Some studies also apply virtual reality (VR) and augmented reality (AR) to assist inspection, for instance, by projecting as-design 3D models on the real components for damage estimation (Karaaslan, Bagci, & Catbas, 2019). Moreover, as NDTs are sensitive to environment factors, e.g. light and temperature, studies tend to apply correlation and regression methods to investigate the relationships between environment factors and data quality. The results can help propose optimal procedures (e.g. the best frequency and timing) to use NDTs (Dai, Rashidi, Brilakis, & Vela, 2013).

Furthermore, some studies propose methods to assess data quality. For SHM, the assessment depends on measuring data package loss and the signal-noise ratio during transmission (Glisic, Yarnold, Moon, & Aktan, 2014). Some studies treat reliability of sensors as random variables which can be estimated by experts or quantified and updated using methods such as Bayesian updating (Cappello, Zonta, Pozzi, Glisic, & Zandonini, 2015; Kim & Frangopol, 2011; Zonta, Glisic, & Adriaenssens, 2014). Methods to automatically detect poor quality data and isolated fault sensors are also developed, including statistical approaches, e.g. Bayesian linear model and statistical hypothesis testing (Huang, Yi, & Li, 2017), and ML models e.g. ANN, clustering, and abnormality detection (Zhou, Yi, & Chen, 2017).

As for data collected by NDTs, the quality can be assessed based on the frequency of missing data, abnormal data values, and conformance of recorded data to standards in inspection results (McClure & Daniell, 2010). Another popular topic is the probability of detection (POD) which determines the possibility that NDTs can correctly detect the minimum damages (e.g. crack width) required by the O&M teams. POD can be expressed using distributions (e.g. cumulative lognormal and exponential) based on comparing NDTs results with true damage data, considering false positives and negatives (Soliman, Frangopol, & Kim, 2013; Wang, Hao, & Fu, 2012).

4.3. Data storage and organisation

4.3.1. Data formats

To properly store data, both raw data and meaningful data need certain formats which regulate how data are described in computer memory. Table 7 lists dominant formats in the literature. Different data types need different formats. Most formats (e.g. .csv and .pdf) are common and traditional formats, while new formats, such as .hdf5 and .ifc, are being increasingly adopted. The former is good at handling massive SHM data. The latter can encode geometric (e.g. geometry of components and damages) as well as some semantic data (e.g. condition indexes) in BIM (Huthwohl, Brilakis, Borrmann, & Sacks, 2018; Zhang, O'Connor, van der Linden, Prakash, & Lynch, 2016).

4.3.2. Data schema and database

Schemas describe how data are structured and related in databases. Many articles adopt traditional relational schema due to the low technical requirements and compatibility with structured data, e.g. sensor readings (Bae, Lee, & Park, 2016; Kurata et al., 2013). In addition, mark-up language schema, e.g. eXtensible mark-up language (XML), is commonly used. XML can encode both structured and non-structured data in a standard and interoperable way and can be extended to suit specific tasks (Jeong, Hou, Lynch, Sohn, & Law, 2019; Zhu et al., 2020). For instance, Jeong, Hou, Lynch, Sohn, and Law (2017) applied the openBrIM schema to describe bridge drawings and inspection reports, and Jeong et al. (2016) used the sensorML schema to improve the organisation of sensor-based data.

With the increasing diversity of data types, a standard and software neutral schema is necessary to integrate and share data efficiently (Costin et al., 2018). Thus, Industrial Foundation Class (IFC), which is based on EXPRESS and STEP language and is originally designed for buildings, has been borrowed in the bridge sector. Continuous efforts are made to extend IFC for bridge projects (Huthwohl et al., 2018; Zhang et al., 2016). The latest IFC schema can encode various data for bridge O&M (e.g. alignments, geometry and structure conditions) and can be converted to XML (i.e. ifcXML). However, DoTs can take different approaches to organise bridge components. For instance, DoTs in the US divide a bridge into components (e.g. girder and deck), whereas DoTs in Japan divide a bridge into finite elements (Masahiro & Takashi, 2013; MLIT, 2015). These requirements can affect the way that inspection data are recorded. Thus, it is crucial to ensure that data schema matches the monitoring and inspection procedures to smoothly integrate the data into the database.

Databases should align with data formats and schemas. As most articles use relational structured data, central relational databases, e.g. Oracle and DB2 are the main options (Bae et al., 2016; Morgenthal et al., 2019; van der Velde, Klatter, & Bakker, 2013). However, to improve data management efficiency, there is an increasing number of studies that apply distributed database architecture where separated databases are created to handle different data formats (Miyamoto & Asano, 2017; Zhang et al., 2016). Another trend is that NoSQL databases (e.g. MongoDB and Apache Cassandra) which have better scalability when handling big data, are emerging to work with .hdf5 files and store massive sensor readings (Jeong et al., 2017, 2019).

4.4. Data sharing

Data sharing is critical so that relevant data can be delivered to the right people at the right time to support bridge O&M applications (Liu & El-Gohary, 2017).

4.4.1. Query and communication

A query extracts data from databases for sharing. The selection of queries relies on databases. Hence, relational

Table 7. Mainstream data formats.

Category	Data type	Mainstream formats
Raw data	Images and videos	.avi, .jpg, .png (Boddupalli et al., 2019; Javadnejad et al., 2017; Shim et al., 2017)
	Infrared data	.gif, .csv, .fpf, .mat (Abu Dabous et al., 2017; Javadnejad et al., 2017)
Meaningful data	Point clouds	.pcd, ASCII (Khaloo et al., 2018; Shim, Dang, Lon, & Jeon, 2019)
	Acoustic and radar data	.csv, ASCII (Abu Dabous et al., 2017; Alsharqawi et al., 2018)
	Structure responses	.xls, .csv, .txt, ASCII, .hdf5, .json (Jeong et al., 2017, 2019; Shim et al., 2017; Zhang et al., 2016)
	Environment data	
	Traffic data	
	Social data	
	Bridge profile data	.dwf and .dxf (drawings) (Abu Dabous et al., 2017), .mov (VR and AR models) (Hilfert & König, 2016; Jauregui, White, Pate, & Woodward, 2005), .rvt and .ifc (3D and BIM models) (Boddupalli et al., 2019; Chan et al., 2016; McGuire et al., 2016)
	IMR&R data	.pdf, .txt (Shim et al., 2019; Shim et al., 2017)
	Damages and defects data	.csv, .jpg, .png (Javadnejad et al., 2017; Shim et al., 2017)

databases apply SQL, MySQL, and PostgreSQL (Feltrin, Meyer, Bischoff, & Motavalli, 2010; Glisic et al., 2014), and NoSQL databases have specific queries, such as the Cassandra query and MongoDB query for Apache Cassandra and MongoDB, respectively (Jeong et al., 2016, 2017). Once extracted, data should be delivered to the right person on time through web, which requires various Internet protocols.

The File Transfer Protocol and Hyper Text Transfer Protocol are widely used and are effective at handling large and small size files, respectively (Koo et al., 2013). As mobile devices are more frequently adopted for inspection, specific protocols (e.g. High-Speed Downlink Packet Access) can be applied to improve communication between mobile devices and servers (Hoult, Fidler, Hill, & Middleton, 2010). Besides, data security is another concern. As such, multiple access control and standard secure shell are common methods to improve security (Zhang et al., 2016; Zhou et al., 2017). Some articles argue that better security depends on asking stakeholders to clarify data ownership and responsibilities during data sharing (van der Velde et al., 2013).

4.4.2. Collaboration platform

Apart from querying and communication, collaboration platforms are also necessary to support collaboration (Sankaran et al., 2016). These platforms provide data interface for all stakeholders and encapsulate functions such as data extraction, analysis, visualisation, and decision-making (Kuckartz & Collier, 2016). The most common digital collaboration platforms are BMS. Common BMS in practice includes the PONTIS developed in Canada (Thompson, Small, Johnson, & Marshall, 1998) and BRIDGIT and BrM developed in US (Hawk & Small, 1998; Modjeski & Masters Inc, 2017). The J-BMS developed by Japan DoTs, which supports expert system and can integrate real-time SHM data with decision-making, is also widely recognised (Miyamoto & Asano, 2018; Miyamoto, Kawamura, & Nakamura, 2000).

Meanwhile, customised BMS is developed and refined continuously by studies and DoTs to support specific requirements or advanced algorithms. In such BMS, modules are usually developed for different functions, e.g. damage estimation, decision-making, and visualisation, and then

data requirements, level of details (LOD), and data flows can be defined between modules to facilitate data exchange (Feltrin et al., 2010; Javadnejad, Gillins, Higgins, & Gillins, 2017). BMS is often built on common web-service architectures, e.g. service-oriented architecture and representational state transfer. Application Program Interfaces are also extensively adopted to allow users to call encapsulated functions in BMS (Jeong et al., 2019).

Although BMS can facilitate data sharing, collaboration issues still remain. One major reason is that DoTs can use different software tools which lead to databases with different schemas. Thus, data in separated BMS cannot be exchanged, which forms the 'data island' problem (Bradley et al., 2016). As such, articles start to develop platforms that support software neutral schemas, especially BIM and ArcGIS, to facilitate data exchange (Javadnejad et al., 2017; Shim et al., 2017). In addition, a few articles build collaboration platforms based on ontology and semantic web techniques (SWT) that link separated data sources and disambiguate information from different stakeholders. Such platforms are useful when dealing with semantic and unstructured data (e.g. inspection results) (Costin et al., 2018). For instance, Liu and El-Gohary (2017) built the BridgeOnto to increase the searching efficiency of IMR&R information in BMS; and in practice, the road agency in Netherland has implemented an ontology based management system to integrate asset data (Luiten et al., 2019; NRA, 2018).

5. Application areas of data-driven decision-making in bridge O&M

The meaningful data can be used in various applications for data-driven bridge O&M. Table 8 shows the distribution of articles of these applications. Evaluating structure conditions is the most common application as condition indexes can reflect structure performance clearly and is required by most DoTs (Modjeski & Masters Inc, 2017). Decision-making also gains considerable research attention as many DoTs need informed decisions to better allocate limited funds (Aukland University, 2015). In addition, there are many studies generating knowledge of bridge structures that are useful for other applications. Finally, evaluating failure

Table 8. Distribution of application areas of bridge O&M data.

Assessment criteria type	Counts
Generating structure knowledge	103
Evaluating failure probability	71
Evaluating structure condition indexes	224
Evaluating life expectancy	55
Making bridge O&M decisions	158

probability and life expectancy are less addressed, probably due to the additional data requirements and more complex computation.

5.1. Structure related knowledge investigation

Based on the O&M data, two types of structure knowledge are commonly investigated.

5.1.1. Structure health model development

Many studies in this field aim to develop mathematical models that can accurately evaluate structure damages usually caused by environment impacts (e.g. corrosion) and dead and live loading. Developing structure health models requires accurate structure responses collected by load testing, SHM, and inspection. The data should be combined with material characteristics of components (e.g. comprehensive stress) and environment data (e.g. chloride diffusion rate) to evaluate damage severity and structure conditions (Conde, Eguia, Stavroulakis, & Granada, 2018; Manafpour, Guler, Radlińska, Rajabipour, & Warn, 2018).

One challenge that these studies face is uncertain model parameters, e.g. crack growth rate and initiation time. Uncertain parameters are usually treated as random variables which are estimated with fuzzy sets (Anoop, Raghuprasad, & Rao, 2012) and statistical distributions, e.g. normal, lognormal, uniform, and triangular distribution (Barone & Frangopol, 2014b; Kim, Ge, & Frangopol, 2019). Experts can be consulted to estimate uncertain parameters (Anoop et al., 2012). Meanwhile, studies apply Bayesian updating and GA to update parameters with new available data collected by sensors and NDTs (Conde et al., 2018; Liu & Fan, 2020). Studies also continue to expand the coverage of these models by developing models for more bridge types, e.g. timber and masonry bridges (Lokuge, Wilson, Tran, & Setunge, 2019; Tena, Todisco, & León, 2018), more components, e.g. mechanical and movable components (Catbas et al., 2014), more damage causes, e.g. alkali-silica reaction (Takahashi, Tanaka, & Maekawa, 2018), scour (Li, He, Li, & Jin, 2017) and earthquake (Navabian & Beskhyroun, 2019), and more environment impacts, e.g. wind, and temperature (Ding, Zhou, Li, & Wang, 2012; Zhang, Chouinard, Power, Tandja M, & Bastien, 2020).

In contrast, IMR&R tasks can fix damages and increase bridge health, which can be preventive and essential. Preventive IMR&R is scheduled regularly to reinforce components before failure, which can be proactive and reactive. The former is performed before damages happen and the latter is performed on damaged components. Essential IMR&R is performed after failure, where failed components

are replaced or completely repaired (Okasha & Frangopol, 2010). Using structure health models reviewed above to analyse damage data before and after IMR&R tasks, some studies develop mechanical models to estimate the effect of various IMR&R actions (Tian & Li, 2016) and techniques, e.g. FRP and glass FRP (Du et al., 2019; Gorski, Stankiewicz, & Tatara, 2018).

5.1.2. Affecting factors investigation

Ambient factors can significantly affect bridge structures, especially traffic (e.g. weight and traffic volume) and environment (e.g. wind, temperature, and water waves). Thus, many studies investigate properties (e.g. statistics and probability density functions) of these factors and the change of these properties overtime, aiming to predict their future values (Watson, Crim, Gurley, & Washburn, 2017; Ye, Yuan, Xi, & Liu, 2018). Studies also investigate the relationships between bridge conditions (e.g. structure responses, modal properties, and damages) and various factors such as bridge profile (e.g. type and location), traffic, and environment.

Instead of evaluating specific damage severity, studies investigating affecting factors often have two aims. One is to develop correlation models between the factors investigated and bridge conditions, which identify potential causes of damages and critical factors (Lou et al., 2017; Tena et al., 2018). The other is to propose algorithms that extract modal properties from structure responses caused by environment factors (Chen, Feng, & Tan, 2006; Feng & Feng, 2016; Magalhaes & Cunha, 2016). Recent articles begin to consider correlations between affecting factors once assumed as independent. For instance, Liu, Zhang, Li, Deng, and Jiang (2018) analysed correlations between fatigue load and load cycle, which can improve damage computation involving these parameters. Methods to investigate affecting factors include correlation analysis, linear or non-linear regression, support vector regression, fitting statistical distributions, extreme value analysis, copula function, and clustering (Asadollahi & Li, 2017; Liu & Fan, 2020; Radovic et al., 2017).

5.2. Failure probability evaluation

In the literature, failure probability can be evaluated with following mechanisms.

5.2.1. Deterioration severity computation

These studies estimate the probability that deterioration (e.g. loss of stiffness) exceeds the limit. The input data can include: 1) bridge profile, e.g. type, age, geometry, and materials characteristics; 2) damage data, e.g. the size of crack; 3) environment data, e.g. chloride concentration and diffusion, temperature, and humidity. Structure health models are common tools to compute deterioration severity. The models are either developed in that study as introduced in Section 5.1, or drawn from well-known researches, e.g. the Paris law for fatigue damages and Fick's second law for corrosion damages.

The failure probability is often evaluated by feeding structure health models with above input data estimated with certain distributions (Kala, 2019). With similar inputs, some studies train ML models (e.g. ANN, SVM, and extreme learning machine) to detect deterioration severity and growth (Firouzi & Rahai, 2013; Hoang, Liao, & Tran, 2018; Huang, 2010). The failure probability can also be directly evaluated using distributions that describe decay of components, where the Weibull and gamma distribution are extensively adopted (Thompson, 2012; van Noortwijk & Frangopol, 2004).

5.2.2. Load computation

Studies in this area estimate the probability that load exceeds the design capacity. Load include dead load of bridge components and live load caused by traffic and environment factors (e.g. wind and temperature). Different causes and components lead to different failure modes, e.g. share, moment, and flexure (Kong & Frangopol, 2003b; Liu & Fan, 2020). The input data for load computation can include: 1) bridge profile; 2) traffic data, e.g. AADT, weight of vehicles, and load cycles; 3) structure responses, e.g. vibration and displacement; and 4) environment data, e.g. temperature and wind speed. Load evaluation can be applied for both normal and extreme situations. The former focuses on evaluating fatigue of structure due to consistent loading, while the latter focuses on the load under extreme scenarios, e.g. earthquake (Guo, Trejo, & Yim, 2015), hurricane (Mondoro, Frangopol, & Soliman, 2017), and extreme heavy vehicles (Bao, Li, Chen, Zhang, & Guo, 2016; Li, Chen, Zhang, Wu, & Wang, 2017; Messervey, Frangopol, & Casciati, 2011).

Load is computed by combining mechanical models and statistical distributions. Mechanical models can transform structure responses to equivalent load (Helmerich et al., 2007) meanwhile distributions can address uncertainties of input data. For instance, many articles apply the Poisson, binomial, exponential, and generalised extreme value distribution to simulate the appearance, load, and extreme load of vehicles, respectively (Chen et al., 2006; Messervey et al., 2011; Watson et al., 2017; Xia et al., 2016). The resultant load is therefore a probabilistic value which can be compared with the design capacity (also known as load-carrying factor or safety index) to estimate the failure probability (Mondoro et al., 2017; Xia et al., 2016).

The current trend of evaluating failure probability is to continuously increase capacity and efficiency of models by combining them with various analytic methods. For instance, Turkan, Hong, Laflamme, and Puri (2018) combines wavelet functions with ANN to separate points clouds into low- and high-resolution surface to improve evaluation of crack severity; Cheng, Cao, and Wu (2015) used optimisation algorithms to calibrate hyperparameters of ML models; and Liu and Fan (2020) considered the correlations between components and failure modes to improve the objectivity of computation.

5.3. Structure condition indexes evaluation

5.3.1. Current condition evaluation

Condition indexes can take three forms with increasing complexity. Binary indexes are the most straightforward form, indicating if damages happen and the structure is out of service. Input data for evaluating binary indexes usually include detected damages (e.g. crack and delamination size) and structure responses (including modal properties derived from the responses). Then, binary indexes can be directly computed by comparing input data with certain thresholds (Sun & Sun, 2018).

Unlike physical damages (e.g. cracks) that are easy to localise and visualise in input data, structure responses are collected by remote sensors and may contain noise and errors due to pre-processing and transmission. Thus, studies propose various methods to identify existence and location of damages objectively and accurately, including statistical approaches, e.g. regression, hypothesis testing, and regularisation (Pan & Yu, 2019), and supervised ML models, e.g. SVM, and ANN (Pan, Azimi, Yan, & Lin, 2018). Unsupervised ML, e.g. k-means clustering, can also be applied to detect damages or categorise damages so that the thresholds are determined more objectively (Omar, Nehdi, & Zayed, 2018). In some cases, control charts are used to facilitate data analysis by enhanced visualisation of the mean, upper, and lower limits of responses (Chen & Durango-Cohen, 2015).

Merely using binary indexes is inadequate for O&M. Thus, many studies adopt discrete indexes (e.g. 0–9) where a lower number indicates more severe damages. In practice, discrete indexes are commonly estimated by inspectors based on engineering experience and rating guidelines, which include detailed description of damages and corresponding ratings (AASHTO, 2010; Mainroads, 2013; TxDOT, 2020). When NDTs and SHM are used, discrete indexes can be computed by mapping detected damages, abnormal responses, and deterioration computed by structure health models to discrete values. For instance, McGuire, Atadero, Clevenger, and Ozbek (2016) and Abu Dabous et al. (2017) evaluated deck conditions using a piece-wise function, considering the percentage of cracking area; and Sasmal, Ramanjaneyulu, Gopalakrishnan, and Lakshmanan (2006) applied fuzzy inference to map abnormal structure responses to different damage degrees.

Discrete indexes rely on visual damages rather than actual structure performance. In practice, it is possible that components fail suddenly even their discrete indexes are high (van Noortwijk & Frangopol, 2004). As such, many studies propose continuous indexes, and the typical ones are listed in Table 9. Computing continuous indexes needs data of deterioration severity, load, and failure probability introduced before. Besides, complex continuous indexes can be evaluated based on basic ones, for instance, the redundancy and hazard function rely on the availability index.

5.3.2. Condition development and prediction

Condition estimation reflects current structure conditions, while condition development predicts future conditions (i.e.

Table 9. Mainstream continuous indexes.

Index	Explanation
Safety index	The remaining capacity (e.g. load carrying capacity) of a component w.r.t its design or demand value. Similar indexes include the load-carrying factor (Cheung, So, & Zhang, 2012; Somodikova, Lehky, Dolezel, & Novak, 2016).
Severability	A measure reflecting if the bridge can serve users well, considering both structure safety and configuration (e.g. alignment and clearances) w.r.t a hypothetical new structure (Orcesi et al., 2010; Thompson, 2012).
Reliability	A measure that usually reflects the probability that a component does not fail w.r.t certain limit state at specific time. Usually, reliability is computed by passing the failure probability to the inverse standard normal distribution (Bocchini & Frangopol, 2011; Kong & Frangopol, 2003b; Okasha & Frangopol, 2009; Val & Melchers, 1997).
Availability	A measure of the probability that a component does not fail up to specific time, which can be expressed as the integration of the probabilistic density distribution of the survival probability (i.e. one minus failure probability). Similar indexes include the unavailability (Barone & Frangopol, 2014a; Kim & Frangopol, 2011).
Redundancy	The availability of the structure warning before collapse, which is computed using the availability of the weakest component and the entire bridge system (Okasha & Frangopol, 2009, 2010).
Hazards function	A measure that indicates how fast the structure becomes non-functional at each time, which can be expressed as the derivative of the availability w.r.t time (Barone & Frangopol, 2014a).
Durability	A measure to reflect the ability of a component to resist deterioration, which can be expressed as the time that a component takes to deteriorate or become structure deficient. Similar indexes include the longevity (Anoop et al., 2012; Miyamoto et al., 2000).

time-dependent conditions), where both deterioration and improvement due to IMR&R should be considered.

Binary indexes are predicted considering the decline of discrete and continuous indexes below a threshold. For discrete indexes, most prediction methods describe the probability that a component remains in one condition for a certain period, e.g. Markov model, fault tree, decision tree, event tree and Weibull function (Goyal, Whelan, & Cavalline, 2020; Kim, Frangopol, & Soliman, 2013; Setunge, Zhu, Gravina, & Gamage, 2016). However, most models need the probability that an index changes (i.e. transition probability) due to deterioration or IMR&R, which can be estimated by experts (Setunge et al., 2016) or derived statistically. In the latter case, historical conditions are needed, based on which statistical methods, e.g. classification tress, regression (e.g. linear, logistical, and support vector regression), are applied to derive the probability (Goyal et al., 2020; Robelin & Madanat, 2007). Again, ML models, e.g. ANN, can be used to predict discrete indexes with inspection, bridge profile, and environment data (Callow, Lee, Blumenstein, Guan, & Loo, 2013).

For continuous indexes, some of them are already time-dependent (i.e. hazard function, availability, redundancy, and durability). However, these indexes are based on more basic ones (e.g. reliability) which are in turn predicted by estimating failure probability, damages, and load in future time points. Besides, some studies describe a continuous process with uncertain deterioration rate to predict indexes without IMR&R, e.g. bilinear, quadratic, square root, biquadratic, and exponential (Bocchini & Frangopol, 2011, 2013). The rate of different bridge components and under different environment conditions can be determined by experts or derived from historical conditions data using regression approaches (Tserng, Yin, & Chung, 2009).

On the other hand, the effect of IMR&R can include: 1) an instant increase of a condition index, 2) a delay of deterioration initiation, and 3) a decrease of deterioration rate. It is useful to estimate the effect of different IMR&R actions performed on different components, using mechanical models corresponding to certain IMR&R actions and techniques (Gorski et al., 2018; Williams & Higgins, 2008), consulting experts (Adey & Hajdin, 2011), or fitting distributions, e.g. normal, uniform, lognormal, and triangular (Bocchini & Frangopol, 2011; Liu & Frangopol, 2004).

5.4. Life expectancy evaluation

Life expectancy is closely related to the change of failure probability and condition indexes overtime. Thus, the life expectancy is often estimated as the time that the failure probability and a condition index declining below a threshold (Helmerich et al., 2007; Kim & Frangopol, 2017; Kim & Frangopol, 2018; Sabatino, Frangopol, & Dong, 2016).

Given element-wise inspection is increasingly adopted while components can have different condition indexes owing to unique failure modes (Kong & Frangopol, 2003b), more studies tend to evaluate structure conditions, failure probability, and life expectancy of individual components and then integrate the results as an overall index. Some studies take the condition of the weakest and safest component to represent the system condition for series and parallel systems, respectively (Barone, Frangopol, & Soliman, 2014). An alternative is to sum indexes of components considering their relative importance which can be based on repair costs, material vulnerability, and expert's suggestions (Choi & Seo, 2009; Moufti, Zayed, & Dabous, 2014) or on the contribution of components' damage to the bridge failure (Yang, Frangopol, & Neves, 2006).

Many mentioned models and approaches have been adopted by DoTs, including the Markov-based model adopted by the Netherland government and the reliability model adopted by the UK government (van Noortwijk & Frangopol, 2004). In the literature, the current trend is to refine these models or to combine different models to enhance their capacity. For instance, Goyal et al. (2020) applied the nonhomogeneous and semi-Markov model where the transition probability can change overtime. That study also combined Markov with the Cox proportional hazards model to release restrictions of traditional distributions and include more explanatory variables. Furthermore, similar to building structure health models, parameters in these models and distributions can be calibrated and updated by optimising techniques, e.g. Bayesian updating, case-based reasoning, and GA (Callow et al., 2013; Orcesi & Frangopol, 2011a).

5.5 O&M decision-making

Decision-making is a process of making final judgments based on alternatives to attain certain goals (Woldesenbet, 2014). Bridge O&M decision-making can be divided into two groups. The first group determines the planning of IMR&R tasks. The second group determines the tools (e.g. NDTs) to perform IMR&R tasks. The decision-making can require both initial data (Section 4) and computation results from various applications (Section 5.1 – 5.4).

5.5.1. Decision-making of IMR&R tasks planning

Studies in this group can assist decision-making at bridge and network level. However, among these articles, only 40 (25%) of them address network-level decision-making, probably due to the additional data needs in environment, traffic, and social aspects to solve network-level problems.

5.5.1.1. Bridge-level decision-making. These decisions select components and IMR&R tasks to be performed on them, aiming to: 1) minimise life-cycle costs (LCC) or net present value (NPV) of O&M; 2) minimise the delay of IMR&R tasks before damage and failure of components; and 3) maximise structure conditions (Kabir et al., 2014; Kim et al., 2019). Studies in this field focus on preventive and essential maintenance (Kim & Frangopol, 2018; Okasha & Frangopol, 2010), and some researches consider specific IMR&R actions e.g. silane treatment, cathodic protection, and painting of steel (Frangopol & Liu, 2007; Liu & Frangopol, 2004). The most straightforward decision-making mechanism is to assign IMR&R tasks to components based on their conditions (Dinh & Zayed, 2016; McGuire et al., 2016). Hence, the key factors for decision-making are condition indexes. Then, IMR&R tasks are performed when the indexes decline below a threshold (Garavaglia & Sgambi, 2016; McGuire et al., 2016; Miyamoto & Asano, 2017). The importance of components can be considered using methods introduced in Section 5 to ensure that critical components are treated first (Bolar, Tesfamariam, & Sadiq, 2014; Choi & Seo, 2009).

As O&M funds are limited, most studies consider IMR&R costs for decision-making, including: 1) routine repair and inspection costs, 2) failure costs to replace failed components, 3) social costs, e.g. user costs due to increased travel time and distance when the bridge is closed and fatality costs due to bridge failure, and 4) environment costs, e.g. monetised additional carbon emissions and energy consumption owing to detour and IMR&R (Kim & Frangopol, 2018; Qu, Yu, Zhou, Lin, & Wang, 2020; Sabatino, Frangopol, & Dong, 2015). The overall costs can be obtained by weighted summing, which are regarded as the sustainability index in some articles when social and environment costs are included (Lounis & McAllister, 2016; Sabatino et al., 2016).

Cost data can be collected from databases of local authorities and expressed by triangular or normal distributions (Frangopol & Liu, 2007; Neves, Frangopol, & Petcherdchoo, 2006). Moreover, the number and interval of IMR&R tasks in the lifetime are widely considered. Studies can derive histograms of such data and predict their future values (Kong & Frangopol, 2003a). Some studies apply distributions (e.g. lognormal) to predict the happening of IMR&R, where distribution parameters are based on the deterioration rate of components and experts' suggestions (Kim et al., 2013; Kong & Frangopol, 2003b). Inspection quality (i.e. POD) can also be involved using methods reviewed in Section 4.2, which can adjust the possibility to perform IMR&R tasks thus impact overall costs (Orcesi & Frangopol, 2011b). Many researches take risk as the decision-making criterion by multiplying the failure probability of components with costs. As such, IMR&R tasks are performed on components with high risk or low cost, as long as the components are safe (Barone & Frangopol, 2014b; Sabatino et al., 2015; Saydam & Frangopol, 2015).

The goals of O&M are often contradictory, making the decision-making a multi-objectives optimisation problem (MOOP). One common approach to solve the MOOP is to weighted sum objective functions to form a single-objective optimisation problem (Shim, Lee, & Kang, 2017). Meanwhile, optimisation techniques can be employed to find the Pareto fronts, such as grid searching (Gong & Frangopol, 2020), tree-based algorithms (e.g. the decision-tree and event tree) (Choi & Seo, 2009; Orcesi & Frangopol, 2011b; Soliman et al., 2013), integer, linear, and dynamic programming (Liu & Madanat, 2015; Zayed, 2004), and evolutionary algorithms (e.g. various forms of GA) (Kim et al., 2013; Kim et al., 2019; Liu, Hammad, & Itoh, 1997; Okasha & Frangopol, 2009). Recent studies also begin to train deep-learning models to solve the MOOP (Wei, Bao, & Li, 2020). However, there can be multiple points on a Pareto front, thus, some studies aim to find the optimal solution by assigning weights to different O&M goals.

Above weights required for decision-making can be determined by subjective and objective methods. Typical subjective methods include directly asking decision-makers or experts to assign weights or applying the Utility Theory and Cumulative Prospect Theory to assign weights according to attitudes of decision-makers of possible outcomes and

respective probabilities (Abu Dabous & Alkass, 2011; Gong & Frangopol, 2020; Sabatino et al., 2016). Therefore, risk-taking decision-makers can give more weights to reducing costs than risk-aversion ones. In contrast, objective methods usually determine weights based on standard deviations and correlations of objective values of the Pareto optimal solutions (Kim et al., 2019).

5.5.1.2. Network-level decision-making. The condition of individual bridges can affect the road network, because the close or failure of one bridge can cause adjacent bridges out of service (Liu & Frangopol, 2006). Network-level decision-making needs to prioritise bridges and then schedule IMR&R tasks for them. The decision-making approaches are similar to those for bridge-level decisions. However, the goals can include minimising overall costs and maintenance delay of maintaining all bridges and maximising network performance, such as reducing total travelling time and distance and increasing network connectivity (Bocchini & Frangopol, 2013; Frangopol & Bocchini, 2012). Therefore, factors considered for decision-making should be adjusted.

Specifically, the condition index of the overall bridge rather than individual components, is adopted (Yadollahi, Abd Majid, & Zin, 2015). Besides, factors reflecting network performance, e.g. the ADT and AADT, are also included (Orcesi & Frangopol, 2011a; Orcesi, Frangopol, & Kim, 2010). Some studies consider network topology (e.g. location and distance between bridges) to estimate correlations between bridges and identify critical bridges. Meanwhile, some studies cluster bridges with similar attributes (e.g. type, location and topology properties) so that IMR&R tasks can be planned with minimum moving distance between multiple bridges (Shim & Lee, 2017; Tsai & Yang, 2004). Finally, as more objectives are included when solving the MOOP, studies employ methods (e.g. the dominance relation-based objective reduction approach) to distinguish essential and redundant objectives and increase computation efficiency (Kim & Frangopol, 2017; Kim & Frangopol, 2018).

Despite of above research achievements, making informed IMR&R decisions in practice is still challenging since it needs extensive data and alignment of objectives between different agencies. Thus, most DoTs make decisions in a time-dependent and damage-dependent manner. The former retrofits components after a particular period since the last IMR&R task or replaces components when their designed age is reached. The latter performs IMR&R tasks when damages are identified (Mainroads, 2013; TxDOT, 2020). The most common factors considered for practical decision-making are O&M costs and structure conditions. However, other considerations are increasingly being included. For instance, the US DoTs identified sixteen risks to be evaluated for decision-making (Modjeski & Masters Inc, 2017). Some European countries (e.g. Germany and Norway) have considered maintenance urgency based on not only structure safety but also traffic safety and aesthetics (NCHRP, 2007).

5.5.2. Decision-making of O&M tools selection

These decisions are made based on in-depth investigation of performance, benefits, and costs of O&M tools (mainly SHM and NDTs). Many researches compare the results produced by O&M tools with ground truth, using different indicators for different tools. For SHM, common indicators include accuracy (or error rate) of sensor readings and modal assurance criterion that measures the experimental and derived modal properties (Zhou, Zhang, Yi, Jiang, & Pan, 2017). The ground truth data are often structure responses previously collected in load testing (Jang et al., 2010; Li, Wu, & Shi, 2019). For NDTs, studies usually compare accuracy and efficiency of NDTs when collecting data. For instance, Dai et al. (2013) measured differences of 3D models created by photo- and video-grammetry and laser scanning with results collected by a total station. Zárate, Caicedo, Yu, and Ziehl (2012) measured differences between concrete delamination detected by GPR and laboratory tests on specimens. Subjective criteria (e.g. commercial availability and easiness of use) can be defined. Then, experts' opinions are used to assign weights to such criteria to compute a single score for different NDTs (Hesse, Atadero, & Ozbek, 2017; Vaghefi et al., 2012).

The costs and benefits of O&M tools and materials are also widely investigated. The costs can include initial purchasing and deployment costs and long-term costs, e.g. costs for maintenance and personnel training (Dai et al., 2013; Orcesi & Frangopol, 2011b). Cost analysis is often accompanied with investigation of hard or soft benefits (Carrion, Quintana, & Crespo, 2017a). Hard benefits can be monetised, and several studies propose the concept of the value of information (VoI) to quantify the benefits. These studies assume that SHM and NDTs can provide rich and timely information of structure conditions, which reduces uncertainties of decision-making therefore saves long-term costs. For instance, a major rehabilitation can be avoided as early damages are detected by SHM (FHWA, 2014).

The common method to estimate VoI is to apply Bayesian updating to update the failure probability based on SHM or NDTs data and then compute the difference of economic indicators (e.g. LCC, NPV, and the internal rate of return) when making decisions with and without the information. The tools should only be applied if their costs are less than VoI (Giordano, Prendergast, & Limongelli, 2020; Zonta et al., 2014). In contrast, soft benefits cannot be monetised. Thus, their values rely on subjective appreciation. Such benefits can include increased safety, quality and confidence of O&M staff, gained knowledge of bridge structures, and improved public perception (Agdas et al., 2016; Carrion et al., 2017a; Ford, Cox, Hagar, & Kirkman, 2012). However, given the difficulty to quantify soft benefits, they are often estimated by experts using survey or interview (Ford et al., 2012) or demonstrated by case studies (Carrion et al., 2017a, Carrion, Quintana, & Crespo, 2017b).

Despite of the diverse applications, above articles share some similarities. First, given the uncertainty and lack of data in computation, fitting statistical distributions and drawing upon opinions of experts and decision-makers are

common methods. Based on the fitted distributions, the Monte Carlo simulation is extensively applied to simulate probability density functions and estimate results (Cappello et al., 2015; Garavaglia & Sgambi, 2016; Takahashi et al., 2018). As for subjective opinions, once they are collected in surveys or interviews, they are often absorbed by methods such as fuzzy logic and reasoning, analytic hierarchy process, and quality function deployment (Alsharqawi, Zayed, & Abu Dabous, 2018; Miyamoto & Asano, 2017; Yadollahi et al., 2015).

Besides, Finite Element (FE) models are the most common approach to embed data and mechanical models into software, which automates computation while simulate and visualise damages, modal properties, and deterioration processes (Miyamoto & Yabe, 2012; Sun, Sun, & Chen, 2016; Sun & Sun, 2018). FE models are usually created according to bridge profile data, e.g. drawings and 3D models. However, initial FE models have discrepancies with reality. Therefore, they should be updated before being used for FE analysis, using actual SHM data (Yu & Ou, 2017) or bridge profile collected by NDTs (e.g. LiDAR and LDV) (Conde et al., 2018; Dai et al., 2014). The updating process can be improved by sensitivity analysis (Park, Kim, & Jongchil, 2012) and optimising algorithms, e.g. GA, which can find optimal parameters that minimise the difference between reality and FE models (Huang, Huang, & Pilakoutas, 2018; Wang, Xu, Sun, & Zhu, 2018).

Finally, given the large number of analytic models, some researches tend to compare their performance to facilitate selection. The principle is to compare the outputs of different models (e.g. predicted condition indexes and optimised IMR&R plans) with the ground truth in previous studies or reports (Black, Brint, & Brailsford, 2005; Elbehairy, Elbeltagi, Hegazy, & Soudki, 2006; van Noortwijk & Frangopol, 2004). For this purpose, a few researches propose the credibility index to reflect model reliability, which can be estimated by comparing errors and uncertainties in two candidate models (Garavaglia & Sgambi, 2015).

6. Challenges and future research

This paper reviews mainstream data for bridge O&M, important application areas using these data, and issues of data management. To this end, several challenges are discussed.

Data-driven decision-making includes data collection, storage, sharing, analysis, and decision-making. As mentioned in Section 1, there are already reviews of bridge O&M data collection techniques and analytic algorithms, while the discussion of O&M data management is less covered. However, such issues (e.g. data needs, storage, and sharing) are critical to enable the entire decision-making process (Auckland University, 2015; London DoT, 2013). Thus, this paper focuses on revealing challenges and research opportunities of data-driven bridge O&M from a data management perspective.

First, there is a lack of comprehensive definition of data needs (e.g. data types, descriptions, and LoD). As Costin

et al. (2018) stated, a prerequisite of applying ICTs in transportation infrastructure is a clear definition of data needs. Some studies point out that a challenge to be tackled is the consistent data definitions and LoD in infrastructure projects (Deng, Cheng, & Anumba, 2016; Tolmer, Castaing, Diab, & Morand, 2017). As reviewed in Section 4.2, some studies have addressed data needs for bridge O&M (Helmerich et al., 2007; Sacks et al., 2018; Shim et al., 2017). Besides, there are practical manuals specifying detailed data needs for inspection, such as the ASTM D6432 for GPR data and the ASTM C597 for ultrasonic data (Mainroads, 2013; TxDOT, 2020).

Most existing data definitions are restricted to data collection. This is probably due to the relatively mature development of SHM and NDTs for data collection and a number of studies that compare data collection tools (Agdas et al., 2016; Carrion et al., 2017a; Ford et al., 2012; Vaghefi et al., 2012). Thus, these definitions cannot fully support data-driven applications reviewed in Section 5. The industry has also published several practical standards related to data needs for bridge O&M (Auckland University, 2015; London DoT, 2013). However, they are often general and universal, which cannot reveal specific data requirements and LoD for different applications.

Thus, it is recommended that more researches to be conducted to systematically investigate important applications for data-driven bridge O&M decision-making, along with critical algorithms and models needed by these applications. Then, data inputs required by each application can be defined. Moreover, degree of data demands among stakeholders can vary. For example, the O&M team of one bridge may need detailed damage severity, while a DoT managing many bridges may need an overall condition index for each bridge (Glisic et al., 2014). Hence, future studies can propose a data delivery manual not only for data collection but also for other applications, considering LoD of different stakeholders. This review has summarised several mainstream O&M applications and their data needs. Future studies can refine and expand the work to have more comprehensive coverage of applications and detailed definition of data needs.

Second, more researches are needed to quantitatively assess data quality. When detailed data needs are defined, poor quality data can still hinder bridge O&M decision-making. However, as introduced in Section 4.2, quantitative data quality assessment is largely restricted to data collected by sensors and NDTs. For other data, quantitative assessment is either not mentioned or depends on subjective judgement. For instance, McClure and Daniell (2010) determines if O&M costs data is qualified by checking abnormal data manually; and Adey and Hajdin (2011) estimated inspection data quality according to experience of the O&M team.

To address the challenge, studies are encouraged to propose quantitative quality assessment methods for each data inputs defined in the data needs. Although different data can have different quality metrics, such as damaged pixels for image data and lost signals for sensor readings, a

quantitative assessment can generally include metrics in three aspects, i.e. missing values, erroneous values, and conformance to standards in terms of naming convention, units, and scale. Then, a score can be computed to represent data quality by weighted summing the metrics. Furthermore, to reach a balance between data quality and costs and time to collect data, a quality threshold for different data can be setup so that only data reaching the threshold can be fed into relevant applications. The threshold can be determined based on relative importance of bridges, components, and intended applications (Bush, Omenzetter, Henning, & McCarten, 2013).

Third, even data of good quality are obtained, data integration is also challenging. Since massive O&M data are collected, integrating the data into a repository and sharing them in a timely manner is the key for successful O&M. However, for one thing, the industry lacks neutral data formats and schemas to represent heterogeneous data and support data integration (Bradley et al., 2016; Sankaran et al., 2016). As mentioned in Section 4.4, tools for O&M applications are continuously developed, which are often designed in a standalone manner, i.e. they adopt different data formats and data schemas. Thus, many tools lack the capability of integrating data.

There is also a lack of automatic access to data sources. It is common to extract data from external sources (i.e. second-hand data), e.g. retrieving AADT and condition indexes from database of DoTs (Bocchini & Frangopol, 2013; Liu & Frangopol, 2006; Sacks et al., 2018). However, automatic access to external sources is not well established, possibly because these sources are often managed by different stakeholders. As such, many analytic methods cannot be implemented in practice since it is impractical to manually search data and repeat the searching when data are updated. Although BIM can integrate data, it focuses on individual projects, whereas bridge O&M needs to access external sources at the network level. Besides, O&M data can involve subjectivity, which can cause ambiguity between stakeholders and is hard for BIM and BMS to handle. For instance, the terms ratings, scores, and indexes can be used by different stakeholders to search the same structure condition data, and incorrect data can be extracted if the ambiguity is not addressed (Liu & El-Gohary, 2017).

Although studies are beginning to apply .ifc format and extend schemas (e.g. XML and IFC) for bridge O&M, they mainly encode geometry and structured data and only recently begin to cover semantic and unstructured data (e.g. inspection results) (Huthwohl et al., 2018; Zhu, Wang, Wang, Wu, & Kim, 2019). However, many data types are not covered, especially those that are usually not included in digital models (e.g. environment and traffic data) or those that are difficult to represent by a structured format (e.g. topology of bridges). As discussed in Section 5, such data can be critical for O&M applications. Hence, the lack of neutral formats and schemas can hinder these applications if needed data are not integrated to enable relevant functions. This can also interfere data sharing when one (e.g. decision-maker) attempts to combine data from different

databases, which can cause suboptimal and delayed O&M decisions.

To address the challenge, it is critical for both the academia and the industry to keep extending and implementing advanced schemas (e.g. IFC and XML) and formats (e.g. .ifc) to cover more aspects of bridge O&M. The other option is to develop novel SWT schemas, such as Resources Description Framework (RDF) and Labelled Property Graph (LPG), which are good at handling unstructured and semantic data (Bradley et al., 2016; Costin et al., 2018).

It is also necessary to develop automatic access to separated data sources. One solution is to develop data parsers corresponding to different schemas. The parsers can interpret data schemas and extract meaningful data automatically, e.g. extracting data between label pairs of XML. Such approaches have been applied to integrate BIM and GIS information in building, road, and utility projects (Cheng & Deng, 2015; Deng et al., 2016), hence can be borrowed to integrate bridge O&M data. The Linked Data technique is another option, which is based on RDF and LPG schema and can develop connections between data sources (Bradley et al., 2016). Besides, to minimise ambiguity, a common knowledge base (e.g. ontologies) can be developed to encode data so that stakeholders can search for data with their preferable expressions. The idea is mentioned in several studies and has been adopted by the Netherland Highway authority, but the application is limited to information searching (Liu & El-Gohary, 2017; Luiten et al., 2019; NRA, 2018). Therefore, more researches related to Linked Data and ontologies are needed to enable information merging, disambiguation, and integration for different O&M applications.

Finally, there is a lack of consideration on operational issues of data-driven O&M applications. Applying ICTs to support decision-making will cause change of working procedures and responsibilities and roles of stakeholders and onsite staff (Sankaran et al., 2016; van der Velde et al., 2013). Meanwhile, the need of ensuring data security while maintaining interoperability will increase along with the increase of data volume and demands for data exchange (Mlynarski & Hu, 2016). However, efforts addressing such issues are inadequate. There are studies and guidelines proposing advice of deploying SHM and NDTs (Dai et al., 2013; FHWA, 2018; TxDOT, 2020; Wa, 2013; Watase et al., 2015). However, the main attention is again given to data collection. When it comes to specific applications, most efforts focus on technical details, e.g. improving algorithms and adding functions of applications and collaboration platforms (Glisic et al., 2014; Kuckartz & Collier, 2016; McClure & Daniell, 2010; van der Velde et al., 2013).

Thus, it is suggested future efforts could develop operational standards for bridge O&M applications and set roles and responsibilities so that O&M teams can know who should search, receive, and analyse data whereas who should maintain and share the outputs (Bradley et al., 2016). To keep the balance between data interoperability and security, templates of execution plan or contracts can be developed, which outline data ownership and rights of accessing and

using data among stakeholders (Mlynarski & Hu, 2016). This template can be detailed by O&M teams to be used in specific projects, similar to the BIM execution plan that deals with issues of data liability and ownership in construction projects (Costin et al., 2018). Owing to the subjective nature of operational issues, reviewing successful projects can be a suitable method for relevant studies, which can reveal how ICTs can be used to support data-driven decision-making (van der Velde et al., 2013). It is also important to train and educate bridge O&M stakeholders and staff so that they become familiar with ICTs and data-driven applications. This can not only increase quality of application outputs, but also overcome institutional barriers, e.g. the resistances or lack of resources to change (Costin et al., 2018; Hu, Chong, & Wang, 2019).

7. Conclusion

Bridges are critical infrastructure, and informed decision-making is essential for bridge O&M. Along with the development of ICTs and analytic methods, data-driven bridge O&M decision-making has become a trend. However, little knowledge exists to provide the bridge industry and academia with a full picture of the latest progress, challenges, and trends of research achievements. Thus, this paper presents a comprehensive review of existing researches in this area.

485 journal articles were reviewed. A general picture of the articles is given by illustrating distributions of articles by time, region, and journal. Then, this paper reviews mainstream data types, data management, and applications of data-driven bridge O&M decision-making. Both raw and meaningful data are covered, along with the methods to covert raw data into meaningful data. For O&M applications, the current research focus is on generating structure related knowledge, evaluating failure probability and structure conditions, and making decisions about IMR&R tasks and tools. Mainstream algorithms and techniques adopted in these applications are summarised. Key data management components that support data-driven O&M process are reviewed, including data needs, quality, formats, schemas and sharing. State-of-art practices mentioned in both articles and practical documents are also introduced. Based on the review, four challenges related to data management are identified, i.e. poor definition of data needs, lack of methods to assess data quality, lack of data integration, and inadequate consideration of operational issues. Then, future research opportunities to address these challenges are discussed.

The contribution of this review is twofold. First, it reveals the up-to-date picture of data-driven bridge O&M decision-making for academia and industry stakeholders. Meanwhile, the identified challenges and opportunities can facilitate further studies to develop more cost-effective and efficient and ICTs for bridge O&M. Second, given the industry is facing difficulties to handle overwhelming data and various application tools, O&M teams can use the review to identify necessary data and suitable applications that align with their

needs. On the other hand, despite that practices are covered, this review still focuses on researches. In the future, another study focusing on practices and real projects will be carried out. Cross-comparison between researches and practices will be made so that mode insight can be provided to help the industry to transform towards data-driven O&M decision-making.

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